CR23532 Cyber and Digital Forensics – Lab Record

Oct 19, 2026 · @Danny

Record Details

Field	Details
Name	Daniel Joshua C S
Register Number	241901020
Department	Computer Science and Engineering (Cyber Security)
Course	CR23532 – Cyber and Digital Forensics
Regulation	2023
Year / Semester	III Year / V Semester
Institution	Rajalakshmi Engineering College

Note on screenshots: Every screenshot slot below is marked *[Screenshot X.Y]*. Capture each one on your own system while doing the experiment, so the paths, user name and timestamps are your own.

Experiment 1 – Study of Computer Forensics and Tools Used for Forensic Investigation

AIM:

To learn the fundamentals of computer forensics, understand the stages of a forensic investigation, and study the tools commonly used to carry one out.

SOFTWARE REQUIREMENTS:

- Operating System: Windows / Linux
- Forensic tools: Autopsy, FTK Imager, The Sleuth Kit and related utilities

THEORY:

Computer forensics (digital forensics) is the discipline of finding, gathering, protecting, studying and interpreting evidence stored on computers and other digital devices. It supports the investigation of cybercrime, security breaches, misuse of systems and unauthorised access.

An examiner looks at storage media, operating system records, application logs, browser data, deleted content and similar sources to find material that is relevant to a case.

The central goal is to recover and interpret evidence without altering it, so that the findings remain reliable and can be presented with confidence.

STAGES OF A DIGITAL FORENSIC INVESTIGATION:

1. **Identification** – Locating possible evidence sources such as hard disks, pen drives, mobile phones, PCs and log files.
2. **Collection** – Gathering the relevant data using accepted forensic methods and tools.
3. **Preservation** – Protecting the evidence from change or damage, usually by working on a forensic image and recording hash values to prove integrity.
4. **Examination** – Searching the collected data for relevant files, metadata, deleted items and other artifacts.
5. **Analysis** – Linking the extracted information together to reconstruct events and identify suspicious activity.
6. **Presentation** – Writing up the methods, observations and conclusions in a report fit for technical or legal review.

FORENSIC TOOLS STUDIED:

A. EnCase Forensic – A commercial forensic suite used to acquire, examine and analyse evidence from computers and storage media.

[Screenshot 1.1: EnCase Forensic main window]

B. Forensic Toolkit (FTK) – An investigation suite that provides indexed searching, file analysis and review of collected evidence.

C. The Sleuth Kit (TSK) – A free, open-source set of command-line programs for analysing disk images and file-system structures and recovering data from them.

D. Autopsy – A graphical front end built on The Sleuth Kit. It supports disk image analysis, deleted file recovery, keyword search and artifact review.

[Screenshot 1.2: Autopsy case window]

E. FTK Imager – An imaging utility used to create and preview forensic images of storage devices and browse the files and folders inside them.

[Screenshot 1.3: FTK Imager evidence tree and file list]

APPLICATIONS OF COMPUTER FORENSICS:

1. **Cybercrime investigation** – Studying evidence linked to hacking, malware and other attacks.
2. **Data recovery** – Retrieving deleted or damaged files where possible.
3. **Incident response** – Finding the cause and scope of a security incident.
4. **Workplace misconduct cases** – Reviewing digital artifacts during authorised internal inquiries.
5. **Legal proceedings** – Preserving and analysing evidence that may be produced in court.
6. **Network investigation** – Reviewing network logs to trace suspicious connections.

PROCEDURE:

1. Read the basic concepts of computer and digital forensics.
2. List and understand the stages of a forensic investigation.
3. Study the commercial and open-source forensic tools available.
4. Learn the purpose and main features of FTK Imager, Autopsy and The Sleuth Kit.
5. Study where digital forensics is applied, especially in cybercrime and data recovery.
6. Record the observations.

OBSERVATION:

The study explained what computer forensics is and why digital evidence matters in cybercrime cases. Each tool was reviewed for its purpose and features. A forensic investigation was seen to follow a fixed sequence: identification, collection, preservation, examination, analysis and presentation.

RESULT:

Thus, the study of computer forensics and the tools used for forensic investigation was completed successfully.

Experiment 2 – Live Forensics Case Investigation Using Autopsy

AIM:

To carry out a digital forensic investigation in Autopsy by creating a case, adding a data source, processing the evidence and reviewing the artifacts the tool produces.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Autopsy Digital Forensic Platform
- Evidence: a forensic/disk image or a set of logical files

- Hardware: PC with enough free storage and RAM

THEORY:

Autopsy is a graphical forensic platform that runs The Sleuth Kit and other modules underneath. It lets an examiner browse files, search keywords, recover deleted content and review user-activity artifacts from an evidence source.

All work is organised inside a **case**, which stores the case details and the evidence attached to it. A **data source** such as a disk image or a folder of logical files is added to the case. Autopsy then runs the chosen **ingest modules** over the source and lists what they find.

The output helps the examiner identify important files, metadata, user accounts, web activity and other artifacts.

PROCEDURE:

Step 1: Launch Autopsy

1. Download Autopsy from its official site and install it.
2. Open the application.
3. Click **New Case** on the welcome screen.

[Screenshot 2.1: Autopsy welcome screen]

Step 2: Create the case

1. Type a case name (for example, `DJ_Autopsy_Case`).
2. Choose the base folder where the case files will be saved.
3. Click **Next**.

[Screenshot 2.2: New Case Information – case name and base directory]

Step 3: Fill in the optional information

1. Enter a case number (for example, `DF-02`).
2. Enter the examiner name: **Daniel Joshua C S**.
3. Add a short note describing the purpose of the case.
4. Click **Finish**.

[Screenshot 2.3: Optional Information with your examiner name]

Step 4: Add the data source

1. Once the case opens, choose to add a data source.
2. Pick the source type – Disk Image or VM File, Local Disk, or Logical Files.
3. Browse to the evidence image or folder.

4. Select it and continue.

[Screenshot 2.4: Select Data Source Type]

Step 5: Configure the ingest modules

1. Go through the list of ingest modules.
2. Tick the modules needed (Recent Activity, Hash Lookup, File Type Identification, Keyword Search, etc.).
3. Click **Next**.
4. Confirm the settings to finish adding the source.

[Screenshot 2.5: Configure Ingest window]

Step 6: Process the data source

1. Complete the Add Data Source wizard.
2. Let Autopsy analyse the evidence.
3. Wait until processing is complete.
4. Review the results in the case window.

[Screenshot 2.6: "Data source has been added" message]

Step 7: Examine the artifacts

Browse the File Views, Data Artifacts and Analysis Results trees. Select a file to see its metadata – MIME type, size, modified/accessed/created times, MD5 and SHA-256. The artifacts shown depend on the evidence and the modules selected.

[Screenshot 2.7: Listing view with file metadata of a selected file]

VERIFICATION OF EVIDENCE INTEGRITY USING SHA-256:

After the analysis, the hash of the evidence image was recomputed in Windows PowerShell to confirm it had not changed:

```
Get-FileHash "<path to your evidence image>" -Algorithm SHA256
```

[Screenshot 2.8: PowerShell Get-FileHash output]

OBSERVATION:

A new case was created in Autopsy and an evidence source was added to it. The selected ingest modules processed the source and produced artifacts for review. The interface listed the files and related information found, and what appeared depended on the contents of the evidence. The SHA-256 value confirmed the integrity of the evidence.

RESULT:

Thus, a digital forensic investigation was performed using Autopsy by creating a case, adding a data source, processing the evidence and examining the resulting artifacts.

Experiment 3 – Recover Deleted Files Using Forensic Tools

AIM:

To recover a deleted file from a forensic image using forensic tools and confirm that the recovered data matches the original.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- FTK Imager
- Recover My Files (or a similar recovery tool)
- A test USB drive or other test evidence source

THEORY:

Deleting a file usually does not wipe its data from the disk. Most file systems only mark the space it occupied as free. Until new data is written over that space, part or all of the file can still be recovered.

Forensic tools allow an examiner to scan a device or its image for such deleted content. FTK Imager is used to acquire and view the forensic image, and a recovery tool then searches that image for deleted files.

Working on an image keeps the original device untouched, because the image is an exact copy of the source at the time of acquisition.

PROCEDURE:

Step 1: Create a test file

1. Open Notepad.
2. Type a few lines of sample text (case reference, project name, your name **Daniel Joshua C S**, today's date, some keywords).
3. Save it as `deleted_secret.txt` on the test USB drive.

[Screenshot 3.1: Contents of deleted_secret.txt in Notepad]

Step 2: Delete the test file

1. Find `deleted_secret.txt`.
2. Right-click it and choose **Delete**.
3. Check that it no longer appears in its folder.

[Screenshot 3.2: File being deleted / Recycle Bin view]

Step 3: Image the drive with FTK Imager

1. Launch FTK Imager.
2. Go to **File → Create Disk Image**.
3. Select the source type (Physical Drive).
4. Choose the test USB drive that held the file.
5. Enter the case number, evidence number and examiner name.
6. Choose the destination folder and image format (E01).
7. Start imaging and wait for it to finish, including verification.

[Screenshot 3.3: Creating Image progress window]

[Screenshot 3.4: Drive/Image Verify Results showing MD5 and SHA1 match]

[Screenshot 3.5: Image Summary with acquisition start/finish times]

Step 4: Recover and verify the file

1. Load the image into the recovery tool and scan for deleted files.
2. Recover `deleted_secret.txt` to a separate folder.
3. Open the recovered file and compare it with the text typed in Step 1.

[Screenshot 3.6: Recovered folder listing]

OBSERVATION:

A sample text file was created on a test drive and then deleted. The drive was imaged with FTK Imager and the image hashes were verified. The image was scanned with a recovery tool to check whether the deleted file and its contents could be retrieved. Recovery depended on the file system, how the image was taken, and whether the space had been overwritten.

RESULT:

Thus, deleted files were recovered from a forensic image using forensic tools, and the recovered contents were verified against the original.

Experiment 4 – Find the Last Connected USB on the System (USB Forensics)

AIM:

To list the USB devices that have been connected to a computer and examine their details using the USBDeview utility.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: USBDeview (NirSoft)
- A computer on which USB devices have been plugged in earlier

THEORY:

USB forensics studies the traces Windows keeps about USB devices plugged into a system. These traces help an examiner prove that a particular device was used on a machine and when.

USBDeview reads these records and shows both currently attached and previously attached USB devices. For each device it can show:

- Device name and description
- Device type (mass storage, HID, video, etc.)
- Serial number, if the device reports one
- Date and time the device was first and last connected
- Vendor ID (VID)
- Product ID (PID)

These fields help build a timeline of USB usage during an investigation.

PROCEDURE:

1. Download USBDeview and run it on the Windows PC.
2. Let it load the list of USB devices recorded on the system.
3. Go through the entries for devices that are connected now and those used earlier.
4. Note the name and description of the device of interest.
5. Check its serial number, Vendor ID and Product ID.
6. Check the Registry Time / connect time values shown for it.
7. Write down the relevant details.
8. Take screenshots of the full list and of the selected device's Properties window.

[Screenshot 4.1: USBDeview main list of devices]

[Screenshot 4.2: Properties window of a selected device]

OBSERVATION:

USBDeview listed the USB devices attached to the computer, including ones that had been used in the past. The device details, such as type, VID, PID, serial number and connection times, were examined to identify the USB connection records.

RESULT:

Thus, USBDeview was used to examine USB connection records and view the device details for forensic investigation.

Experiment 5 – View the Last Activity of the PC

AIM:

To review recent user actions and system events recorded on a Windows computer using LastActivityView.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: LastActivityView (NirSoft)

THEORY:

LastActivityView gathers information from several places in Windows (Prefetch files, registry keys, event logs, recent-file lists and so on) and arranges it as a single log of actions on the computer.

This gives an examiner a quick view of activities such as programs run, files and folders opened, system start-up and shutdown, and network connections. What appears depends on which records are still present on the system.

PROCEDURE:

1. Download LastActivityView and run it on the Windows PC.
2. Let it scan the available system records.
3. Look through the activity log in the main window.
4. Note the details shown for each entry – Action Time, Description, Filename, Full Path, More Information and Data Source.
5. Pick out the entries that relate to recent activity of interest.
6. Record the findings.
7. Take screenshots of the log and of one entry's Properties window.

[Screenshot 5.1: LastActivityView activity log]

[Screenshot 5.2: Properties of a selected entry]

OBSERVATION:

LastActivityView showed the recorded activities and events on the Windows computer, with the time and source of each. These entries helped in understanding what had recently been done on the system.

RESULT:

Thus, LastActivityView was used to examine records of recent computer activity and system events for digital forensic investigation.

Experiment 6 – Extraction of Browser Artifacts

AIM:

To extract and examine browser artifacts – cached files and browsing history – using ChromeCacheView and ChromeHistoryView.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Web Browser: Google Chrome
- Tools: ChromeCacheView and ChromeHistoryView (NirSoft)

THEORY:

Browser artifacts are traces a browser leaves on disk while it is used – cache, history, cookies, downloads and so on. They often reveal which sites a user visited and when, which makes them valuable in investigations.

ChromeCacheView reads Chrome's cache folder and lists the cached items, with fields such as URL, content type, file name, file size and last-accessed time.

ChromeHistoryView reads Chrome's History database and lists visited pages, with fields such as URL, page title, visit time, visit count, referrer and profile.

Together they let an examiner reconstruct a user's web activity.

PROCEDURE:

A. Examination using ChromeCacheView

1. Download ChromeCacheView and run it.
2. Let it find and load the Chrome cache folder.
3. Go through the cached items in the main window.
4. Note the URL, content type, file name, size and access time of each item.
5. Select the entries relevant to the investigation.
6. Take a screenshot of the cache list.

[Screenshot 6.1: ChromeCacheView cache list]

B. Examination using ChromeHistoryView

1. Download ChromeHistoryView and run it.
2. Let it find and load the Chrome History file.

3. Go through the history records shown.
4. Note the URL, title, visit time and visit count.
5. Open the Properties of one record to see its referrer, transition type and source History file.
6. Take screenshots of the history list and the Properties window.

[Screenshot 6.2: ChromeHistoryView history list]

[Screenshot 6.3: Properties of a selected history record]

OBSERVATION:

S. No.	Tool	Artifact Examined	Observation
1	ChromeCacheView	Cached files	Details of cached items were listed where available
2	ChromeHistoryView	Browsing history	Visited website records were listed where available

RESULT:

Thus, browser artifacts were examined using ChromeCacheView and ChromeHistoryView, and the cached file details and browsing history records were extracted and analysed for digital forensic investigation.

Experiment 7 – Comparison of Two Files for Forensic Investigation Using Compare It

AIM:

To compare two files with Compare It and identify the differences between their contents for forensic investigation.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Compare It
- Two sample text files

THEORY:

File comparison shows exactly where two files differ. In forensics it is used to check whether a document has been tampered with, or to see what changed between two versions of the same file.

Compare It places both files side by side and marks differences in colour, so added, removed and changed lines can be spotted at once. This helps an examiner judge how and where a file was altered.

PROCEDURE:

1. Install Compare It on the Windows PC.
2. Prepare two text files with mostly the same content and a few changes (for example, `original_file.txt` and `modified_file.txt` with different case reference, date, prepared-by name and password hint).
3. Open Compare It.
4. Choose the first file and click **Open**.
5. Choose the second file and click **Open**.
6. Let the tool show both files side by side.
7. Look at the highlighted lines to find the differences.
8. Note which lines were added, removed or changed.
9. Record the observations and take screenshots of the result.

[Screenshot 7.1: Side-by-side comparison with highlighted differences]

[Screenshot 7.2: Differences-only / report view]

OBSERVATION:

Compare It displayed both files next to each other and highlighted every line that differed, making the modified fields easy to identify.

RESULT:

Thus, two files were compared using Compare It, and the differences in their contents were identified for forensic examination.

Experiment 8 – Collecting Email Evidence in a Victim PC

AIM:

To gather email-related evidence from a computer by capturing its RAM with DumpIt and analysing the memory image for relevant information.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: DumpIt
- Memory analysis tool: bulk_extractor (or the tool specified in the lab)

THEORY:

Email forensics is the process of locating and studying email-related data on a system to understand who communicated with whom and when.

While a computer is running, its RAM holds temporary data from open applications, including webmail pages in a browser. **DumpIt** is a small utility that copies the entire physical memory of a Windows machine into a single dump file.

The dump is then processed by an analysis tool such as **bulk_extractor**, which scans it and writes out features it finds – email addresses, URLs, domains and so on – into separate text files. The dump must be stored safely and hashed to keep the evidence reliable.

PROCEDURE:

A. Capturing RAM using DumpIt

1. Download DumpIt onto the Windows PC.
2. Run `DumpIt.exe` as Administrator.
3. Read the prompt showing the destination path and machine details.
4. Type **y** to start the acquisition.
5. Wait until the capture completes.
6. Go to the folder where the `.dmp` file was saved.
7. Note the dump file name and its size.

[Screenshot 8.1: DumpIt console during acquisition]

B. Examining email-related evidence

1. Load the memory dump into bulk_extractor and select an output folder.
2. Enable the required scanners (including **email**) and run the scan.
3. Open `email.txt` to view the email addresses found in memory.
4. Open `email_histogram.txt` to see how often each address appears.
5. Note the relevant findings.
6. Keep the dump and the output files for documentation.

[Screenshot 8.2: bulk_extractor run settings]

[Screenshot 8.3: Output folder listing]

[Screenshot 8.4: Contents of email.txt / email_histogram.txt]

OBSERVATION:

The RAM of the system was captured with DumpIt into a dump file. Scanning the dump produced feature files, and the email-related output listed the addresses present in memory along with their frequency.

RESULT:

Thus, email-related evidence was collected from a computer by acquiring its RAM with DumpIt and examining the memory dump.

Experiment 9 – Hiding and Extracting a Text File Behind an Image/Audio File Using Command Prompt

AIM:

To hide a text file behind an image file and an audio file using Command Prompt, using WinRAR to prepare the archive for the audio method, and to recover the hidden file.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Command Prompt (CMD)
- WinRAR (for the audio method)
- Image file (`B.jpeg`), audio file (`sound.mp3`)
- Text files (`A.txt` , `secret.txt`)

THEORY:

File concatenation joins two files end to end to make one output file. Image viewers and media players read only the part of the file they understand and ignore the extra bytes at the end, so the output still opens normally while carrying hidden data.

In the image method, a text file is appended directly to a JPEG. In the audio method, the text file is first zipped with WinRAR and the ZIP is appended to an MP3. Because ZIP tools look for the archive structure at the end of a file, WinRAR can still open the combined MP3 and extract the hidden file.

Forensic examiners look for such extra data by inspecting file sizes, trailing bytes and file signatures.

PROCEDURE:

A. Hiding a text file behind an image file using CMD

1. Create a folder named `hiding_text_in_image` on the Desktop.
2. Copy the image `B.jpeg` and the text file `A.txt` into it.
3. Open Command Prompt.
4. Move into the folder:

```
cd hiding_text_in_image
```

5. Run the command below to join the files:

```
copy /b B.jpeg + A.txt C.jpeg
```

Where:

- `B.jpeg` – original image
- `A.txt` – text file to be hidden
- `C.jpeg` – combined output file
- `/b` – copy in binary mode

6. Wait for the "1 file(s) copied" message.
7. Confirm that `C.jpeg` exists and opens as a normal image.
8. Open `C.jpeg` in Notepad and scroll to the end to see the hidden text.

[Screenshot 9.1: Folder with A.txt and B.jpeg]

[Screenshot 9.2: CMD output of the copy /b command]

[Screenshot 9.3: C.jpeg opened in Notepad showing the text at the end]

B. Hiding a text file behind an audio file using WinRAR and CMD

1. Create `secret.txt` in Notepad with some sample text and save it.
2. Right-click `secret.txt` and use WinRAR to make `secret.zip`.
3. Keep `sound.mp3` and `secret.zip` in the same folder (`Steganography`).
4. Open Command Prompt and move into that folder.
5. Run:

```
copy /b sound.mp3+secret.zip newfile.mp3
```

Where:

- `sound.mp3` – original audio
- `secret.zip` – archive holding the text file
- `newfile.mp3` – combined output file
- `/b` – copy in binary mode

6. Wait for the command to finish and confirm `newfile.mp3` is created and still plays.
7. Optionally open `newfile.mp3` in Notepad to see the archive data at the end.
8. To recover, open a copy of `newfile.mp3` in WinRAR, extract it to a new folder, and check that `secret.txt` is intact.

[Screenshot 9.4: CMD output of the audio copy command]

[Screenshot 9.5: Folder showing newfile.mp3 size vs sound.mp3]

[Screenshot 9.6: WinRAR extraction window]

[Screenshot 9.7: Recovered files in the extraction folder]

RESULT:

Thus, a text file was hidden behind an image file using Command Prompt to produce `C.jpeg`, and a ZIP archive containing a text file was hidden behind an audio file using WinRAR and Command Prompt to produce `newfile.mp3`, from which the hidden file was successfully extracted.

Experiment 10 – Extract EXIF Data from Image Files Using ExifReader

AIM:

To extract and study the EXIF (Exchangeable Image File Format) metadata of an image file using ExifReader.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: ExifReader
- A JPEG/JPG image taken with a phone or camera

THEORY:

EXIF is a standard for storing metadata inside digital photos. Cameras and smartphones write it automatically when a picture is taken. It can include:

- **Camera make and model** – the device that took the photo.
- **Date and time** – when the photo was captured.
- **Camera settings** – shutter speed, aperture (F-number), ISO and exposure.
- **Flash** – whether the flash fired.
- **Image dimensions** – width and height in pixels.
- **GPS data** – location coordinates, if location tagging was on.

ExifReader is a Windows program that opens an image and lists all of its EXIF tags. Examiners use this data to link a photo to a device, a time and sometimes a place.

PROCEDURE:

1. Download ExifReader on the Windows PC.
2. Run `ExifRead.exe`.
3. Click **Open**.
4. Browse to the image to be examined.

5. Select a JPEG/JPG file and open it.
6. Check that the image preview appears.
7. Go through the EXIF tags listed.
8. Note details such as make, model, date/time, dimensions, exposure time, F-number, ISO and flash.
9. Record the relevant metadata.
10. Take a screenshot of the image with its EXIF information.

[Screenshot 10.1: ExifReader showing the image and its EXIF tags]

OBSERVATION:

ExifReader displayed the chosen image together with its EXIF metadata, including the device details, capture date and time, and camera settings stored in the file.

RESULT:

Thus, the EXIF metadata of an image file was extracted and examined using ExifReader.